QUBO Formulation: Mutual-Information Sensor Placement Problem

Mutual-Information Sensor Placement Problem

Problem Statement. Given a set of N candidate sensor locations, the goal is to select exactly K locations to maximize the total information score while accounting for pairwise redundancy. Each location $i \in \{0, \ldots, N-1\}$ is associated with an information score I_i, and each distinct pair of locations $\{i, j\}$ has a redundancy penalty R_{ij} that quantifies the loss in mutual information when both sensors are simultaneously active.

Decision Variables. For each candidate location $i \in \{0, \ldots, N-1\}$, introduce a binary decision variable $x_i \in \{0, 1\}$. The variable $x_i = 1$ indicates that location i is selected, and $x_i = 0$ otherwise.

QUBO Derivation. Step 1 — The original maximization objective is $\max \sum_{i=0}^{N-1} I_i x_i - \sum_{0 \le i < j \le N-1} R_{ij} x_i x_j$. Converting to a minimization problem for the QUBO framework yields:

$$\min \left(- \sum_{i=0}^{N-1} I_i x_i + \sum_{0 \le i < j \le N-1} R_{ij} x_i x_j \right). \tag{1}$$

Step 2 — The cardinality constraint requires exactly K sensors to be selected:

$$\sum_{i=0}^{N-1} x_i = K. \tag{2}$$

Step 3 — We enforce this constraint via a quadratic penalty term with weight $\lambda > 0$:

$$P(x) = \lambda \left(\sum_{i=0}^{N-1} x_i - K \right)^2 \tag{3}$$

$$= \lambda \left(\sum_{i=0}^{N-1} x_i^2 - 2K \sum_{i=0}^{N-1} x_i + K^2 + \sum_{i \neq j} x_i x_j \right). \tag{4}$$

Using the idempotent property $x_i^2 = x_i$ for binary variables, and noting that $\sum_{i \neq j} x_i x_j = 2 \sum_{0 \leq i < j \leq N-1} x_i x_j$, we expand and collect terms:

$$P(x) = \lambda \sum_{i=0}^{N-1} (1 - 2K) x_i + 2\lambda \sum_{0 \leq i < j \leq N-1} x_i x_j + \lambda K^2. \tag{5}$$

Step 4 — The total QUBO Hamiltonian $H(x)$ is the sum of the negated objective and the penalty term:

$$H(x) = -\sum_{i=0}^{N-1} I_i x_i + \sum_{0 \leq i < j \leq N-1} R_{ij} x_i x_j + \lambda \sum_{i=0}^{N-1} (1 - 2K) x_i + 2\lambda \sum_{0 \leq i < j \leq N-1} x_i x_j + \lambda K^2. \tag{6}$$

Step 5 — Grouping linear and quadratic terms yields the standard QUBO form $H(x) = \sum_i Q_{i,i} x_i + \sum_{i<j} Q_{i,j} x_i x_j + c$. The Q matrix entries and offset are given in closed form as:

$$Q_{i,i} = -I_i + \lambda(1 - 2K), \qquad \text{for } i \in \{0, \ldots, N-1\}, \tag{7}$$

$$Q_{i,j} = R_{ij} + 2\lambda, \qquad \text{for } 0 \leq i < j \leq N-1, \tag{8}$$

$$c = \lambda K^2. \tag{9}$$

For $i > j$, we set $Q_{i,j} = Q_{j,i}$ to maintain symmetry.

Penalty Weight Justification. The penalty weight λ must be chosen sufficiently large to ensure that any violation of the cardinality constraint results in an energy strictly greater than the optimal feasible solution. Setting $\lambda > \max_i I_i$ ensures any constraint violation costs more than the maximum possible gain from the objective function. Here $n = N$.

Correctness Argument. Minimizing $H(x)$ over $\{0,1\}^N$ recovers the optimal sensor placement because (i) all feasible solutions satisfying $\sum_i x_i = K$ incur zero penalty, so their energy equals the negated objective value; (ii) any infeasible solution violates the cardinality constraint by at least one unit, incurring a penalty of at least λ. Since $\lambda > \max_i I_i$, the energy of any infeasible solution strictly exceeds that of any feasible solution. Consequently, the global minimum of $H(x)$ must be feasible and corresponds to the configuration that maximizes the original objective.

Illustrative Example. Consider a small instance with $N = 3$ candidate locations and $K = 2$ required sensors. Let the information scores be $I_0 = 10$, $I_1 = 8$, $I_2 = 6$, and the redundancy penalties be $R_{01} = 2$, $R_{02} = 3$, $R_{12} = 1$. We choose $\lambda = 11$, which satisfies $\lambda > \max\{10, 8, 6\}$. Substituting these values into the general formulas yields the concrete Hamiltonian:

$$\begin{aligned} H(x) &= (-10 - 33)x_0 + (-8 - 33)x_1 + (-6 - 33)x_2 \\ &\quad + (2 + 22)x_0 x_1 + (3 + 22)x_0 x_2 + (1 + 22)x_1 x_2 + 44 \\ &= -43x_0 - 41x_1 - 39x_2 + 24x_0 x_1 + 25x_0 x_2 + 23x_1 x_2 + 44. \end{aligned} \tag{10}$$

Evaluating $H(x)$ over the three feasible assignments gives $H(1,1,0) = -16$, $H(1,0,1) = -13$, and $H(0,1,1) = -13$. The global minimum occurs at $x^* = (1,1,0)$ with energy -16, corresponding to selecting locations 0 and 1 for a maximum total information score of $10 + 8 - 2 = 16$.